7.4 LUBRICATION SYSTEM OF I.C. ENGINES :

The friction between surfaces rubbing against each other is reduced by using a lubricant between them. Normally the engine manufacturer recommneds a mineral oil of given viscosity. Oil is classified by the Society of Automobile Engineers (S A E) number. Higher the number thicker is the oil. An SAE 40 oil is more viscous than SAE 30.

7.4.1 Purpose of Lubrication :

- prevention of metal to metal contact so as to reduce friction, wear and tear.
- assisting the cooling system by carrying away heat from various parts such as crown of piston, valve stems, connecting rod etc.
- providing a seal between cylinder walls and piston rings and thus avoid leakage of pressure.
- providing a cushioning effect against vibration and impact
- protection against corrosion
- cleaning the various parts of the engine by carrying away grit and carbon deposits from the working surfaces.

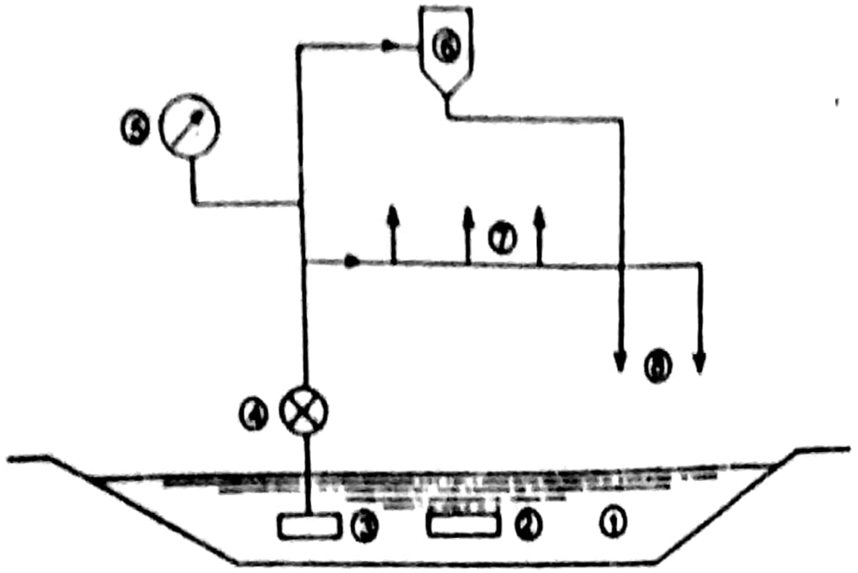
7.4.2 Parts that Require Lubrication :

In an I.C engine following are the parts that require lubrication.

- Crankshaft bearings
- Crankpins
- Small end and big end of connecting rod
- Gudgeon pin bushes
- Inner wall of cylinder
- Piston rings
- Valve operating mechanism
- Timing gear
- Camshaft bearings

7.4.3 Lubricating System :

Fig. 7.21 illustrates the main parts of a lubricating system



1. Oil Sump 2. Oil Screen 3. Oil Pump 4. Pressure Relief
5. Pre. Gauge 6. Oil Filter 7. Oil Gallery 8. Return Line

Fig. 7.21

The main parts of the lubricating system are :

Pump : Driven by the camshaft, the pump forces the oil from the sump to the oil gallery. It may either be kept submerged in the oil or be above the oil level in the sump.

Main Oil Gallery : This runs along the length of the engine. It is the passage through which the lubricating oil from the pump is forced to the various parts of the engine. It is also known as oil header. In some engines oil galleries are drilled in the engine block itself, whereas in some engines steel or copper tubes are used as oil galleries.

Relief Valve : Generally fitted in the gallery, this spring loaded valve opens when the pressure of oil reaches maximum allowed value.

Filter : Besides the gauze screen that prevents the pieces of metal entering the pump, this oil filter cleanses the lubricating oil from foreign matter.

Pressure Gauge : It records the oil pressure in a pressure feed system.

7.4.4 Types of Lubrication System :

There are mainly three types of lubricating systems.

(a) Petroil system

(b) Splash system

(c) Pressure system

(a) Petroil System : This is the simplest form of lubrication and is generally adopted in 2-stroke petrol engines. That is why scooters, moped, motor cycles running on 2-stroke petrol engines adopt this system. In this system lubricating oil is mixed with petrol. Usually 30 ml of oil with 1 litre of petrol is mixed and simultaneously fill the tank. No separate oil pump is provided in the engine.

During engine operation, lubricating oil particles percolate into the bearing surfaces and lubricate them.

The main drawback with this system is that lubricating oil at times does not thoroughly mix with petrol and separates off from it when engine is not run for a considerable period. This leads to clogging of passages in the carburettor resulting in starting trouble. Large deposits of carbon collect in the piston crown, transfer port and exhaust pipe. This results in loss of power and increased fuel consumption. Secondly, depending on quality of lubricating oil, the 1.00 : 0.30 ratio of oils may prove excess or shortage of lubricating oil.

(b) Splash System : In this system oil is splashed over different working parts of an engine. Oil is contained in a through or sump (Fig. 7.22). The big end of connecting rod is provided with a 'spoon or

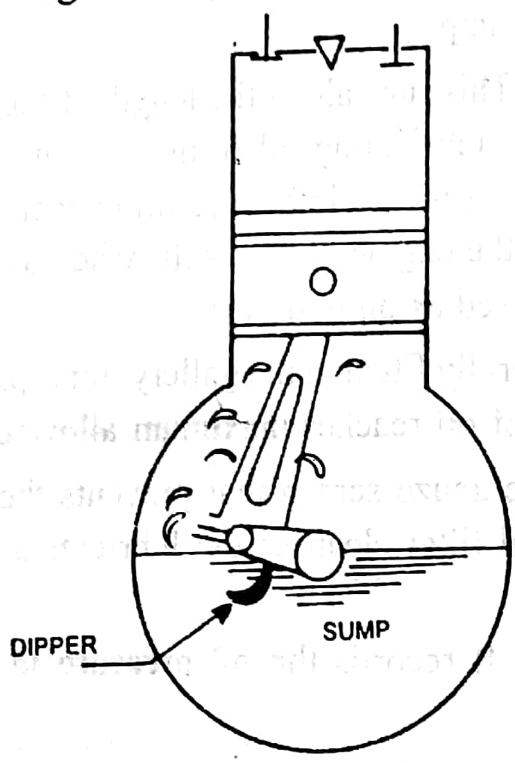


Fig. 7.22 SPLASH LUBRICATION

dipper' or 'scoop'. When the piston is at the bottom of its stroke, the big end of connecting rod and crankpin dip into oil. The dipper picks up oil and as the crankshaft rotates, oil is splashed up due to centrifugal force.

The splashed oil is in the form of a dense mist sprayed into fine particles over surfaces in contact. Small cups are provided close to the bearing of the crank shaft. There are small holes in these cups. The splashed oils fills up these cups from where it is supplied to the bearings.

Oil that is splashed on to cylinder walls speeds well when piston reciprocates while the piston rings scrape the oil and get themselves lubricated. Drops of splashed oil drip from the inner side of the piston and lubricate the gudgeon pin and bearings. The camshaft bearings, valve mechanism and timing gears are also lubricated by splashed oil.

(c) Pressure Feed System : In an I.C engine clearance between mating parts will be less than 0.001 mm. Splashed oil does not have sufficient pressure to get into such a small clearance. In pressure feed system this drawback is overcome by using an oil pump which boosts up pressure and oil is forced through the small gap between mating surfaces.

Fig. 7.23 illustrates the pressure feed system of lubrication.

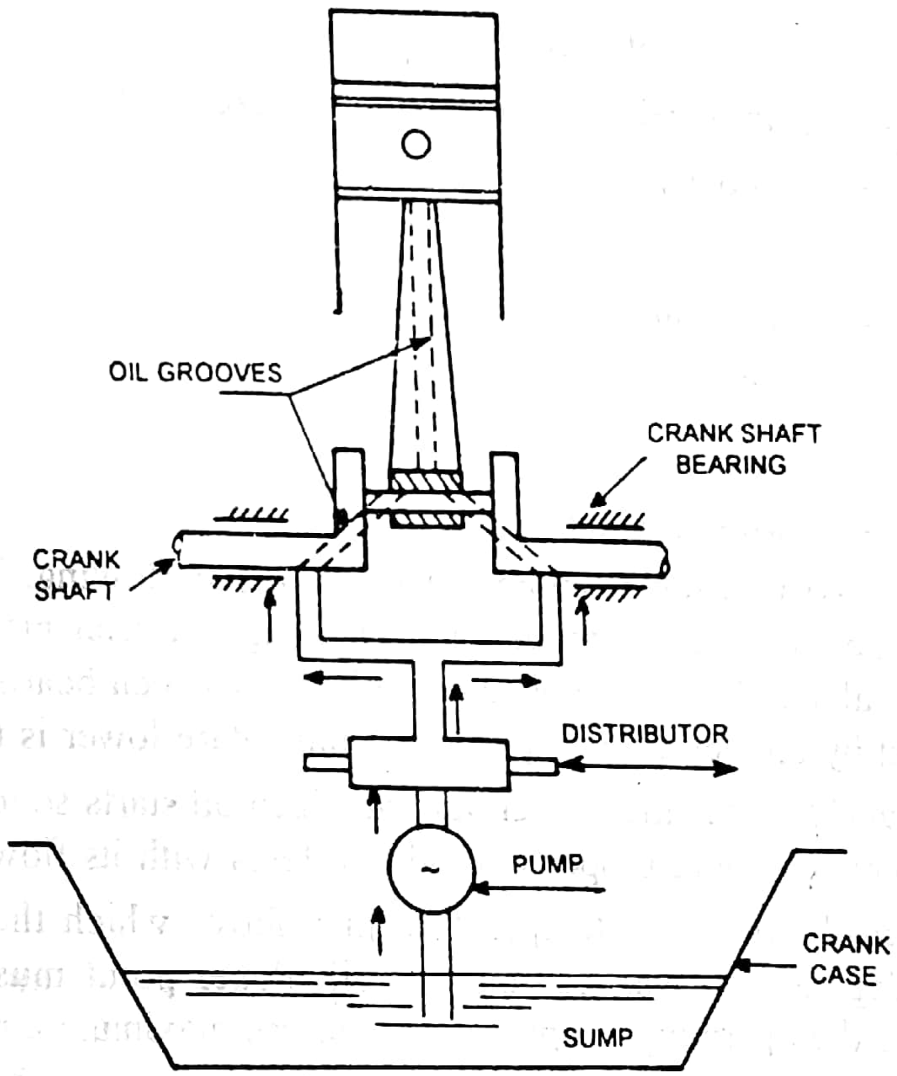


Fig. 7.23 PRESSURE FEED LUBRICATION

Oil pump is operated by a camshaft and is placed in the sump. The pump draws oil from the sump and delivers to the distributor. Pipes leading from distributor carry oil to the various parts of the engine. In fig. 7.23 one such pipe leading from distributor is shown bifurcating and supplying oil to the main bearings of crankshaft. Through small grooves inside the shaft, oil reaches big end of connecting rod. Drilled passages through connecting rod carry oil to piston, gudgeon pin etc. Other pipes leading from distributor supply lubricating oil in the same way to valve mechanism, timing gear etc.

Combined splash and pressure feed system of lubrication may also be employed.